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Effect of Cyanobacterial toxin on germination of wheat seed

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Abstract : A large number of Blue-green Algae produces toxin. The toxin produced by *Microcystis aeruginosa* is known as Microcystin L.R. This toxin is hazardous to aquatic flora and fauna. It is also harmful to crop plants and animals. This toxin inhibits seed germination of several crop plants. In this study effect of microcystin L.R. derived from local strain of *Microcystis aeruginosa* was tested on germination of wheat seed. Concentration of toxin higher than 40% showed inhibition of seed germination.

Keywords: Cynobacterial toxin, germination.

INTRODUCTION

Cyanobacteria are the most common attainder in bloom formation. Majority of them are known to produce potent biotoxins. Cyanotoxin produced by Cyanobactria fall into three broad groups :-

- Cyclic peptides
- Alkaloids
- Lipopoly saccharides

Cyclic peptides are produced by a large number of cyanobacteria. The toxins produced by *Microcystis aeruginosa* are known as microcystin LR. Microcystin is also produced by several other cyanobactria such as *Anabaena, Oscillatoria, Nostoc, Hapalosiphon, Ananbaenopsis,* the other cyclic peptide nodularin is produced by *Nodularia*. The alkaloids includes anatoxin-a, aplysiatoxin, cylandrospermopsins, Lyngbyatoxin-a, saxitoxins. These toxins are produced by *Anabaena, Oscillatoria, Lyngbya, Schizothrix, Cylindrospermum, Cylindrospermopsis, Aphanizomenon* etc. cyclic peptides and alkaloids are more hazardous to aquatic fauna, plants and animals. These causes damage to liver, nervous system and skin in mammals.

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In present study effect of microcystin produced by *Microcystis aeruginosa* was tested on seed germination of wheat.

MATERIALS AND METHODS

Axenic culture from local strain of *Microcystis aeruginosa* was cultured in Jaworski's medium and toxin was extracted.

Extraction of Toxins

Lyopholized cells were mixed within 20 ml of 50% aqueous methanol and sonicated in an ice bath for 10 min in a sonicator. Sonicated suspension was stirred at room temperature for 12h. The suspension was centrifuged (10000 rpm for 15min) and the pellet was re-extracted using 10 ml of the extraction solvent. Extracts was evaporated to dryness in vacuum at 40°C and dissolved in 3.0 ml of 10ml ammonium acetate. After removal of the insoluble materials by centrifugation (10000 rpm for 10 min) an aliquot of the supernatant (2.0 ml) was passed through a C_{18} cartridge. The cartridge was washed with 10 ml water and the toxin eluted with 3ml of methanol.

Five different concentration of toxin were prepared (20%, 40%, 60%, 80%, 100%). 100 seeds of wheat were soacked in each concentration for 24 hours. 100 wheat seeds were shocked in distilled water which were used as control. The seeds were grown in plastic boxes on filter papers shocked with same concentration of toxin. For

control seeds were grown in plastic box on filter papers soacked with distilled water. During germination 4 ml extract of each concentration was added on 3 day intervals to avoid dryness of germinating seeds.

RESULTS

After 12 days seeds germinated. Seedlings were observed. Seedlings of 20% toxin concentration were normal while 40%, 60% and 80% concentration showed poor germination. In 100% concentration seed germination was negligible. Percentage germination of seed is shown in table no.-1.

Table No.-1: Percentage germination of wheat seed in toxin of *Microcystis aeruginosa*

Concentration of extract	Percentage germination of seed	Percentage germination of seed in cuntrol
20%	85	94
40%	70	94
60%	40	94
80%	10	94

CONCLUSION

Percentage of germinated seeds get reduced due to toxins of *M. aeruginosa* which is dependednt on concentration of toxin.

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